

Friday 01/17/2025 section #6 Hint for HW#1

Drag is given by $D = c_D \frac{1}{2} \rho u(h)^2 A_f$ (1)

In HW#1, $c_D = 0.5$, $A_f = A_e$, $\rho = \rho(h) \rightarrow \begin{cases} h < 11 \text{ km} \rightarrow \rho = \rho_0 (\dots) \\ h > 11 \text{ km} \rightarrow \rho = \rho_1 (\dots) \end{cases}$

The key point is that velocity (u) depends on drag, i.e.,

$$u - u_0 = u_{eq} \ln \frac{M_0}{M(t)} - g \cos \theta t - \left(\frac{\bar{D}}{M} \right) t \quad (2)$$

So from (1), $D = D(u)$ & from (2) $u = u(D)$!

In HW#1, assume one (u), compute $\bar{D}$, put $\bar{D}$ in (2), calculate new velocity, compare the new velocity with your assumed value, if did not match, change u & iterate!

Define $|\text{Error}| = \text{Abs} \left(\frac{u(\text{new}) - u(\text{old})}{u(\text{old})} \right) < \frac{\text{epsilon}}{0.01}$

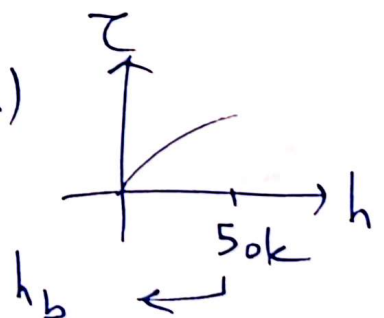
Hint 1 - A similar code for h_b iteration is provided in Discord channel HW#1

Hint 2 - $\frac{\bar{D}}{M} = \frac{\int_0^t \frac{D}{M} dt}{t}$ (use numerical integration)

HW#1 asks for $\tau(h)$

$$\tau = \underbrace{m u_e}_{\text{fixed}} + \underbrace{(p_e - p_a(h))}_{\text{fixed}} \underbrace{A_e}_{\text{fixed}} \rightarrow \tau = \tau(h)$$

(P-1)



The question asks $u_{eq}(h) = \frac{\tau(h) \rightarrow \text{Variable}}{\dot{m} \rightarrow \text{constant}}$ The shape is the same as thrust

" " " $I_{sp}(h) = \frac{u_{eq}(h)}{g(h)} \rightarrow \text{Both changes}$

" " " $t_b = \frac{M_0 - M_b}{\dot{m}} = \text{fixed, the same as Example}$

In the HW, you need to plot $0 < t < t_b$

$$a_0 = \frac{\dot{m} u_{eq}}{M_0} - g, \quad u_{eq}(h=0) \text{ \& } g = g_e \Rightarrow a_0 = \frac{\dot{m} u_{eq}(h=0)}{M_0} - g_e$$

$$a_b = \frac{\dot{m} u_{eq}}{M_b^-} - g \cos \theta - \frac{D}{M_b^-} \quad (M_b^- = M_b^+ = M_b), \text{ given in Question}$$

$h \rightarrow h_b \rightarrow$ compute u_{eq} @ h_b , θ at h_b , D at h_b

You ~~need to~~ have computed h_b in part (a)

Today course: Sounding Rocket, total impulse, revisiting velocity rise E_f^h , h_b , h_{max} , $h(t) \rightarrow \text{Altitude at time } t!$

Sounding Rocket is a type of rocket that carries simple missions in sub-orbital trajectory. A payload is housed in the nose cone, which separates before/during descent.

(P-2)

Total Impulse: $I = \int_0^{t_b} \tau dt = \int_0^{t_b} \dot{m} u_{eq} dt = u_{eq} \int_0^{t_b} \dot{m} dt = M_p u_{eq}$

Revisiting Velocity rise Eqⁿ

start from acceleration Eqⁿ: $\frac{du}{dt} = \frac{\dot{m} u_{eq}}{M} - g \cos \theta - \frac{D}{M}$

$\& \dot{m} = -\frac{dM}{dt} \rightarrow du = -u_{eq} \frac{dM}{M} - g \cos \theta$, Assume $\theta = 0$, $g = g_e$

then integrate $\rightarrow u(t) = -u_{eq} \ln\left(\frac{M(t)}{M_0}\right) - g_e t$

Replace M as a function of time:

$M(t) = M_0 - \dot{m} t$ $\rightarrow$ if $t = 0 \rightarrow M = M_0$
if $t = t_b$ $M = M_0 - \dot{m} t_b = M_b$

$\Rightarrow M(t) = M_0 - \dot{m} t = M_0 - \left(\frac{M_0 - M_b}{t_b}\right) t$ put this M(t) in u(t) Equation

we get:

$u(t) = -u_{eq} \ln(\text{argument}(t)) - g_e t$

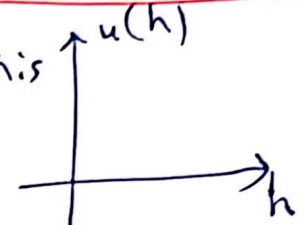
$\text{argument}(t) = \left[1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right]$, $\begin{cases} R = \frac{M_0}{M_b} \\ \frac{1}{R} = \frac{M_b}{M_0} \end{cases}$

This is velocity at time t

(P-3)

In $H \neq 1$, $u_{eq} = u_{eq}(h)$, so we need a relation for h as a function of t . It is given by:

$$h(t) = \frac{u_{eq} t_b R}{R-1} \left[\text{argument}(t) \times \ln(\text{argument}(t)) - \text{argument}(t) + 1 \right] - \frac{1}{2} g_e t^2$$

In $H \neq 1$, you should plot this 

write a loop between $t < t_b$, compute $h(t)$ at any t , then find $u(t)$ at that time

$$t = 0 \rightarrow u = u_0, h = h_0$$

$$t = t_1 \rightarrow u = u_1, h = h_1$$

⋮

plot these data

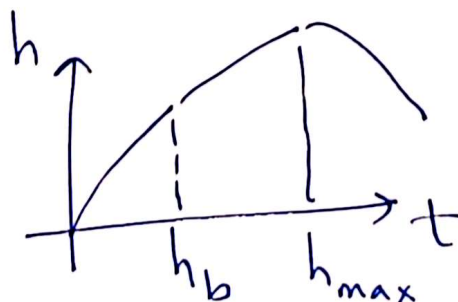
⋮

Burn out altitude

$$h_b = -u_{eq} t_b \frac{\ln R}{R-1} + u_{eq} t_b - g_e \frac{t_b^2}{2}$$

Max altitude

$$h_{max} = -u_{eq} t_b \left[\frac{R \ln R}{R-1} - 1 \right] + \frac{u_{eq}^2 (\ln R)^2}{2 g_e}$$



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(P-8)

h_b & h_{max} depends on u_{eq} & u_{eq} depends on h ,

Assume $h \rightarrow$ compute u_{eq} , put in h_b & h_{max}

if $h_{assumed} = h_{max} \rightarrow$ your assumption is right,
" $= h_b$ otherwise iterate!

code given in Discord for h_b